Database Technologies – Assignment 10 **KD2-Abhishek-96835**

1. Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, Perimeter = 2*(L+W) and Area = L*W.

```
KD2_Abhishek_96835>DELIMITER $$
KD2_Abhishek_96835>
KD2_Abhishek_96835>CREATE PROCEDURE sp_area_rect(p_length INT,p_width INT)
    -> BEGIN
    ->   DECLARE perimeter INT;
    ->   DECLARE area INT;
    ->   SET perimeter=2*(p_length+p_width);
    ->   SET area=p_length*p_width;
    ->   SELECT p_length AS length,p_width AS width, perimeter, area;
    -> END;
    -> $$
Query OK, 0 rows affected (0.01 sec)

KD2_Abhishek_96835>
KD2_Abhishek_96835>DELIMITER ;
KD2_Abhishek_96835>CALL sp_area_rect(21, 5);
+--------+-------+-----------+------+
| length | width | perimeter | area |
+--------+-------+-----------+------+
|     21 |     5 |        52 |  105 |
+--------+-------+-----------+------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```

2. Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```
KD2_Abhishek_96835>CREATE TABLE temp (num INT, squares INT, cubes INT);
Query OK, 0 rows affected (0.03 sec)

KD2_Abhishek_96835>DELIMITER $$
KD2_Abhishek_96835>
KD2_Abhishek_96835>CREATE PROCEDURE sp_square_cube(n_num INT)
    ->
    ->     BEGIN
    ->       DECLARE num INT;
    ->       DECLARE squares INT;
    ->       DECLARE cubes INT;
    ->       SET num=n_num;
    ->       SET squares=num*num;
    ->       SET cubes=num*num*num;
    ->       INSERT INTO temp VALUES(num,squares,cubes);
    ->
    ->      END;
    ->      $$
Query OK, 0 rows affected (0.01 sec)

KD2_Abhishek_96835>DELIMITER ;
KD2_Abhishek_96835>SELECT * FROM temp;
Empty set (0.00 sec)

KD2_Abhishek_96835>CALL sp_square_cube(5);
Query OK, 1 row affected (0.01 sec)

KD2_Abhishek_96835>SELECT * FROM temp;
+------+---------+-------+
| num  | squares | cubes |
+------+---------+-------+
|    5 |      25 |   125 |
+------+---------+-------+
1 row in set (0.00 sec)
```

3. Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are C= (F-32) *5/9 F= 9/5*C + 32

```
KD2_Abhishek_96835>DELIMITER $$
KD2_Abhishek_96835>
KD2_Abhishek_96835>CREATE PROCEDURE sp_temp(num FLOAT, ch CHAR)
    ->
    ->    BEGIN
    ->      DECLARE fa FLOAT;
    ->      DECLARE ce FLOAT;
    ->      IF 'f' =ch THEN
    ->        SET ce=(num-32)*5/9;
    ->      ELSE
    ->    SET fa =9/5*num + 32;
    ->      END IF;
    ->      SELECT fa as Fahrenheit, ce as Celsius;
    ->    END;
    ->
    -> $$
Query OK, 0 rows affected (0.01 sec)

KD2_Abhishek_96835>
KD2_Abhishek_96835>DELIMITER ;
KD2_Abhishek_96835>CALL sp_temp(100,'f');
+------------+---------+
| Fahrenheit | Celsius |
+------------+---------+
|       NULL | 37.7778 |
+------------+---------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

KD2_Abhishek_96835>CALL sp_temp(100,'c');
+------------+---------+
| Fahrenheit | Celsius |
+------------+---------+
|        212 |    NULL |
+------------+---------+
1 row in set (0.00 sec)
```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```
KD2_Abhishek_96835>DELIMITER $$
KD2_Abhishek_96835>
KD2_Abhishek_96835>CREATE PROCEDURE sp_yard(num INT)
    ->
    ->    BEGIN
    ->      DECLARE yard INT;
    ->      DECLARE foot INT;
    ->      DECLARE inches INT;
    ->      DECLARE rem INT;
    ->      SET yard =num/36;
    ->      SET rem=num%36;
    ->      SET foot=rem/12;
    ->      SET inches=rem%12;
    ->      SELECT yard,foot,inches;
    ->
    ->         END;
    ->
    -> $$
Query OK, 0 rows affected (0.01 sec)

KD2_Abhishek_96835>
KD2_Abhishek_96835>DELIMITER ;
KD2_Abhishek_96835>CALL sp_yard(124);
+------+------+--------+
| yard | foot | inches |
+------+------+--------+
|    3 |    1 |      4 |
+------+------+--------+
1 row in set (0.00 sec)
```

5. Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100.

```
KD2_Abhishek_96835>DELIMITER $$
KD2_Abhishek_96835>
KD2_Abhishek_96835>CREATE PROCEDURE sp_product(num1 INT, num2 INT)
    ->
    ->    BEGIN
    ->      DECLARE res INT;
    ->      DECLARE c CHAR;
    ->      SET res=num1*num2;
    ->      IF(res>=100) THEN
    ->         SET c='Y';
    ->      ELSE
    ->         SET c='N';
    ->      END IF;
    ->      SELECT num1,num2,res,c;
    ->    END;
    ->
    -> $$
Query OK, 0 rows affected (0.01 sec)

KD2_Abhishek_96835>
KD2_Abhishek_96835>DELIMITER ;
KD2_Abhishek_96835>CALL sp_product(6,3);
+------+------+------+------+
| num1 | num2 | res  | c    |
+------+------+------+------+
|    6 |    3 |   18 | N    |
+------+------+------+------+
1 row in set (0.00 sec)
```